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Name of Faculty: Parshant Tyagi

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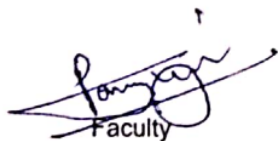
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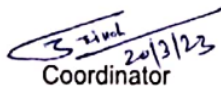
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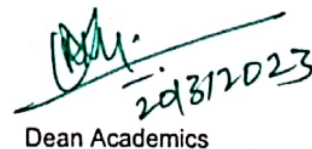
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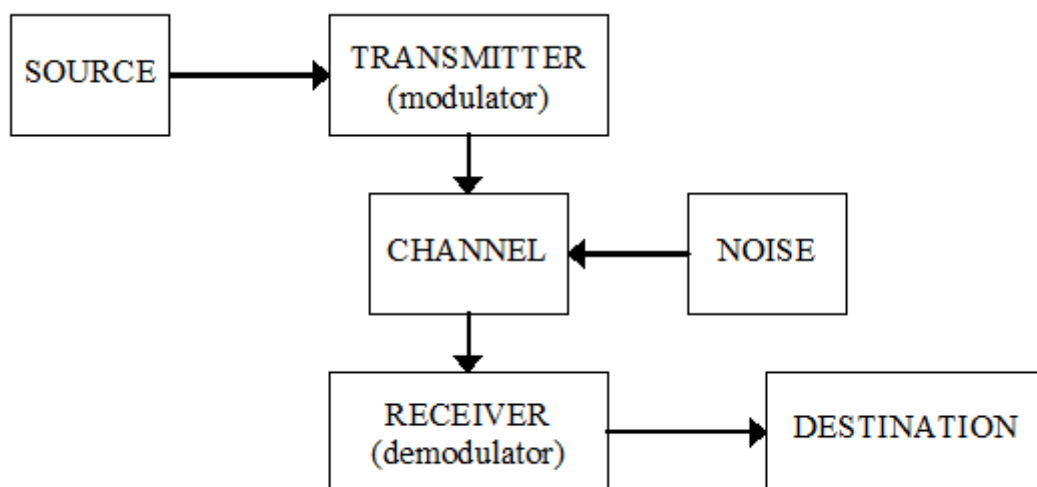
UNIT-5
Communication System
Lecture-1

Introduction to Communication System

Communication is the process by which information is exchanged between individuals through a medium.

Communication can also be defined as the transfer of information from one point in space and time to another point.

The basic block diagram of a communication system is as follows.



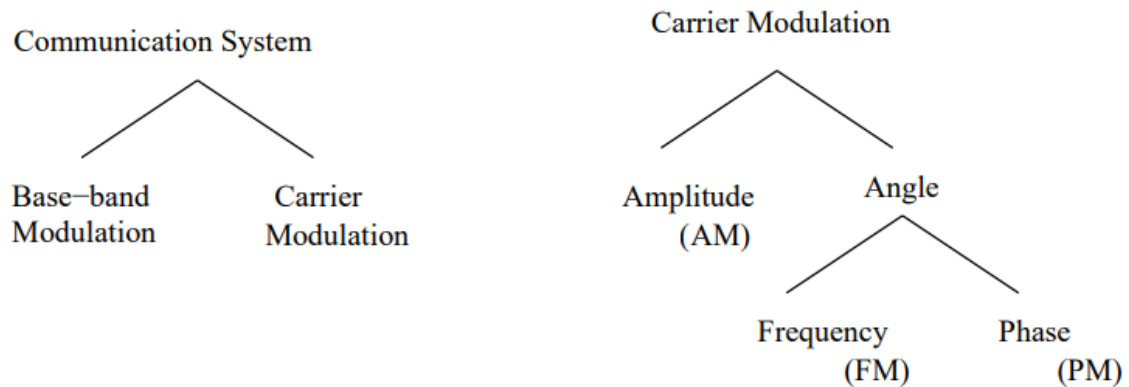
- **Transmitter:** Couples the message into the channel using high frequency signals.
- **Channel:** The medium used for transmission of signals
- **Modulation:** It is the process of shifting the frequency spectrum of a signal to a frequency range in which more efficient transmission can be achieved.
- **Receiver:** Restores the signal to its original form.
- **Demodulation:** It is the process of shifting the frequency spectrum back to the original baseband frequency range and reconstructing the original form.

Modulation:

Modulation is a process that causes a shift in the range of frequencies in a signal.

- Signals that occupy the same range of frequencies can be separated.
- Modulation helps in noise immunity, attenuation - depends on the physical medium.

The below figure shows the different kinds of analog modulation schemes that are available



Modulation is operation performed at the transmitter to achieve efficient and reliable information transmission.

For analog modulation, it is frequency translation method caused by changing the appropriate quantity in a carrier signal.

It involves two waveforms:

- A modulating signal/baseband signal – represents the message.
- A carrier signal – depends on type of modulation.

•Once this information is received, the low frequency information must be removed from the high frequency carrier. •This process is known as “Demodulation”.

Need for Modulation:

- Baseband signals are incompatible for direct transmission over the medium so, modulation is used to convey (baseband) signals from one place to another.
- Allows frequency translation:
 - Frequency Multiplexing
 - Reduce the antenna height
 - Avoids mixing of signals
 - Narrowbanding
- Efficient transmission
- Reduced noise and interference

Types of Modulation:

Three main types of modulations:

Analog Modulation

- **Amplitude modulation**

Example: Double sideband with carrier (DSB-WC), Double- sideband suppressed carrier (DSB-SC), Single sideband suppressed carrier (SSB-SC), vestigial sideband (VSB)

- **Angle modulation (frequency modulation & phase modulation)**

Example: Narrow band frequency modulation (NBFM), Wideband frequency modulation (WBFM), Narrowband phase modulation (NBPM), Wideband phase modulation (WBPM)

Pulse Modulation

- Carrier is a train of pulses
- Example: Pulse Amplitude Modulation (PAM), Pulse width modulation (PWM) , Pulse Position Modulation (PPM)

Digital Modulation

- Modulating signal is analog
 - Example: Pulse Code Modulation (PCM), Delta Modulation (DM), Adaptive Delta Modulation (ADM), Differential Pulse Code Modulation (DPCM), Adaptive Differential Pulse Code Modulation (ADPCM) etc.
- Modulating signal is digital (binary modulation)
 - Example: Amplitude shift keying (ASK), frequency Shift Keying (FSK), Phase Shift Keying (PSK) etc

Frequency Division Multiplexing

Multiplexing is the name given to techniques, which allow more than one message to be transferred via the same communication channel. The channel in this context could be a transmission line, *e.g.* a twisted pair or co-axial cable, a radio system or a fibre optic system *etc.*

FDM is derived from AM techniques in which the signals occupy the same physical 'line' but in different frequency bands. Each signal occupies its own specific band of frequencies all the time, *i.e.* the messages share the channel **bandwidth**.

- FDM – messages occupy **narrow** bandwidth – all the time.

Multiplexing requires that the signals be kept apart so that they do not interfere with each other, and thus they can be separated at the receiving end. This is accomplished by separating the signal either in frequency or time. The technique of separating the signals in frequency is referred to as frequency-division multiplexing (FDM), whereas the technique of separating the signals in time is called time-division multiplexing. In this section, we discuss frequency division multiplexing systems, referred hereafter as FDM.

Fig. 1 shows the block diagram of FDM system. As shown in the Fig. 1, input message signals, assumed to be of the low-pass type are passed through input low-pass filters. This filtering action removes high-frequency components that do not contribute significantly to signal representation but may disturb other message signals that share the common channel. The filtered message signals are then modulated with necessary carrier frequencies with the help of modulators. The most commonly method of modulation in FDM is single sideband modulation, which requires a bandwidth that is approximately equal to that of original message signal. The band pass filters following the modulators are used to restrict the band of each modulated wave to its prescribed range. The outputs of band-pass filters are combined in parallel which form the input to the common channel.

At the receiving end, bandpass filters connected to the common channel separate the message signals on the frequency occupancy basis. Finally, the original message signals are recovered by individual demodulators.

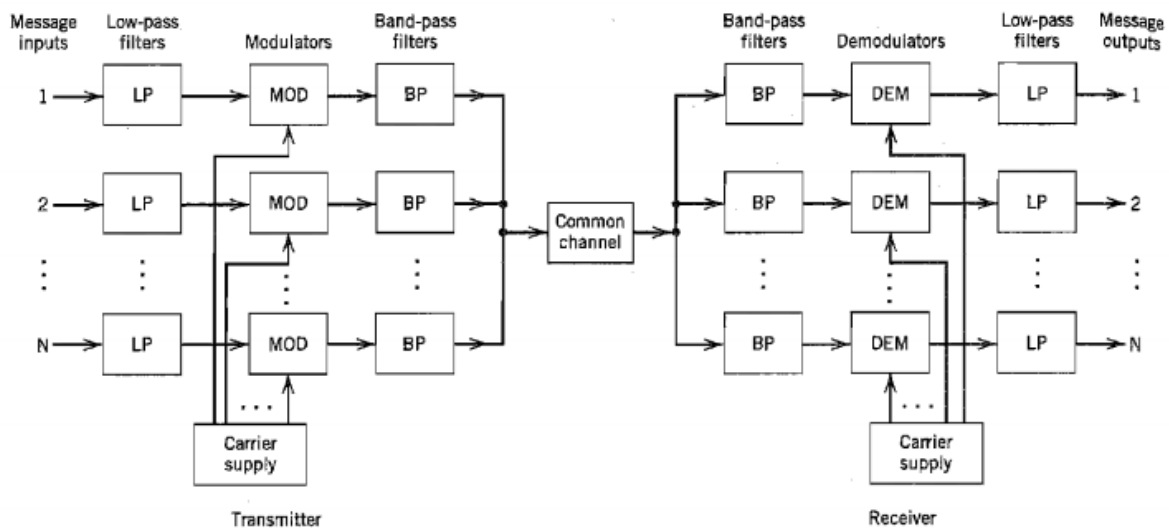


Fig.1. Frequency Division Multiplexing

Amplitude Modulation (AM)

Amplitude Modulation is the process of changing the amplitude of a relatively high frequency carrier signal in accordance with the amplitude of the modulating signal (Information).

The carrier amplitude varied linearly by the modulating signal which usually consists of a range of audio frequencies. The frequency of the carrier is not affected.

Lecture-2 various Types of Modulation

- Application of AM - Radio broadcasting, TV pictures (video), facsimile transmission
- Frequency range for AM - 535 kHz – 1600 kHz
- Bandwidth - 10 kHz

Various forms of Amplitude Modulation

- Conventional Amplitude Modulation (Alternatively known as Full AM or Double Sideband Large carrier modulation (DSBLC) /Double Sideband Full Carrier (DSBFC)
- Double Sideband Suppressed carrier (DSBSC) modulation
- Single Sideband (SSB) modulation
- Vestigial Sideband (VSB) modulation

Time Domain and Frequency Domain Description

It is the process where, the amplitude of the carrier is varied proportional to that of the message signal.

Let $m(t)$ be the base-band signal, $m(t) \longleftrightarrow M(\omega)$ and $c(t)$ be the carrier, $c(t) = A_c \cos(\omega_c t)$. f_c is chosen such that $f_c \gg W$, where W is the maximum frequency component of $m(t)$. The amplitude modulated signal is given by

$$s(t) = A_c [1 + k_a m(t)] \cos(\omega_c t)$$

Fourier Transform on both sides of the above equation

$$S(\omega) = \pi A_c / 2 (\delta(\omega - \omega_c) + \delta(\omega + \omega_c)) + k_a A_c / 2 (M(\omega - \omega_c) + M(\omega + \omega_c))$$

k_a is a constant called amplitude sensitivity.

$k_a m(t) < 1$ and it indicates percentage modulation.

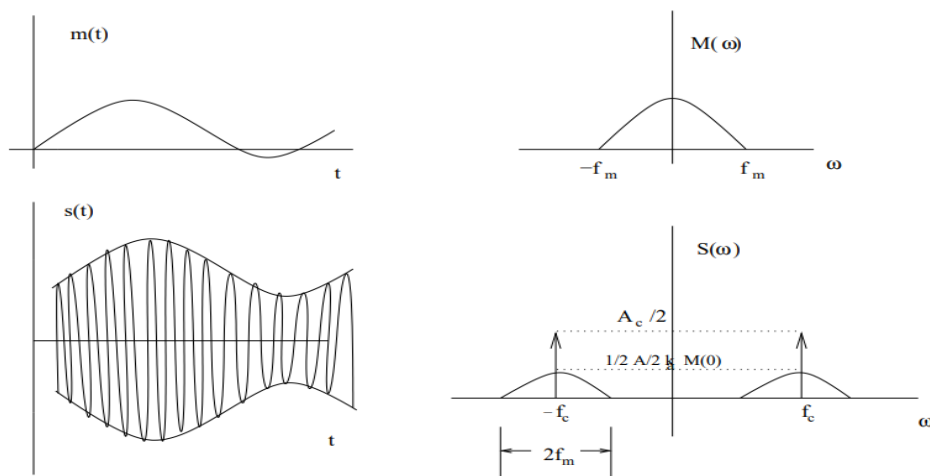


Fig.2. Amplitude modulation in time and frequency domain

Single Tone Modulation:

Consider a modulating wave $m(t)$ that consists of a single tone or single frequency component given by

$$m(t) = A_m \cos(2\pi f_m t) \quad \dots\dots\dots(1)$$

where A_m is peak amplitude of the sinusoidal modulating wave

f_m is the frequency of the sinusoidal modulating wave

Let A_c be the peak amplitude and f_c be the frequency of the high frequency carrier signal. Then the corresponding single-tone AM wave is given by

$$s(t) = A_c [1 + m \cos(2\pi f_m t)] \cos(2\pi f_c t) \quad \dots\dots\dots(2)$$

Let $A_{\max}$ and $A_{\min}$ denote the maximum and minimum values of the envelope of the modulated wave. Then from the above equation (2.12), we get

$$\frac{A_{\max}}{A_{\min}} = \frac{A_c (1 + m)}{A_c (1 - m)}$$
$$m = \frac{A_{\max} - A_{\min}}{A_{\max} + A_{\min}}$$

Expanding the equation (2), we get

$$s(t) = A_c \cos(2\pi f_c t) + \frac{1}{2} m A_c \cos[2\pi(f_c + f_m)t] + \frac{1}{2} m A_c \cos[2\pi(f_c - f_m)t]$$

The Fourier transform of $s(t)$ is obtained as follows.

$$s(f) = \frac{1}{2} A_c [\delta(f - f_c) + \delta(f + f_c)] + \frac{1}{4} m A_c [\delta(f - f_c - f_m) + \delta(f + f_c + f_m)]$$
$$+ \frac{1}{4} m A_c [\delta(f - f_c + f_m) + \delta(f + f_c - f_m)]$$

Thus the spectrum of an AM wave, for the special case of sinusoidal modulation consists of delta functions at $\pm f_c$, $f_c \pm f_m$, and $-f_c \pm f_m$. The spectrum for positive frequencies is as shown in figure

Lecture-3 Power Calculations

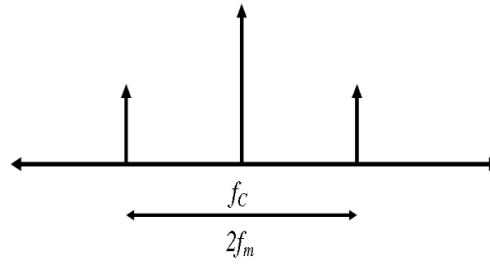


Fig.3. Frequency Domain characteristics of single tone AM

Power relations in AM waves:

Consider the expression for single tone/sinusoidal AM wave

$$s(t) = A_c \cos(2\pi f_c t) + \frac{1}{2} m A_c \cos[2\pi(f_c + f_m)t] + \frac{1}{2} m A_c \cos[2\pi(f_c - f_m)t] \quad \text{.....(1)}$$

This expression contains three components. They are carrier component, upper side band and lower side band. Therefore Average power of the AM wave is sum of these three components.

Therefore the total power in the amplitude modulated wave is given by

$$P_t = \frac{V_{car}^2}{R} + \frac{V_{LSB}^2}{R} + \frac{V_{USB}^2}{R} \quad \text{.....(2)}$$

Where all the voltages are rms values and R is the resistance, in which the power is dissipated.

$$P_C = \frac{V_{car}^2}{R} = \frac{\left(\frac{A_c}{\sqrt{2}}\right)^2}{R} = \frac{A_c^2}{2R}$$

$$P_{LSB} = \frac{V_{LSB}^2}{R} = \left(\frac{mA_c}{2\sqrt{2}}\right)^2 \frac{1}{R} = \frac{m^2 A_c^2}{8R} = \frac{m^2}{4} P_c$$

$$P_{USB} = \frac{V_{USB}^2}{R} = \left(\frac{mA_c}{2\sqrt{2}}\right)^2 \frac{1}{R} = \frac{m^2 A_c^2}{8R} = \frac{m^2}{4} P_c$$

Therefore total average power is given by

$$\begin{aligned}
 P_t &= P_c + P_{LSB} + P_{USB} \\
 P_t &= P_c + \frac{m^2}{4} P_c + \frac{m^2}{4} P_c \\
 P_t &= P_c \left(1 + \frac{m^2}{4} + \frac{m^2}{4} \right) \\
 P_t &= P_c \left(1 + \frac{m^2}{2} \right) \quad \dots\dots\dots(3)
 \end{aligned}$$

The ratio of total side band power to the total power in the modulated wave is given by

$$\begin{aligned}
 \frac{P_{SB}}{P_t} &= \frac{P_c (m^2 / 2)}{P_c (1 + m^2 / 2)} \\
 \frac{P_{SB}}{P_t} &= \frac{m^2}{2 + m^2} \quad \dots\dots\dots(4)
 \end{aligned}$$

This ratio is called the efficiency of AM system

Generation of AM waves:

Two basic amplitude modulation principles are discussed. They are square law modulation and switching modulator.

Square Law Modulator

When the output of a device is not directly proportional to input throughout the operation, the device is said to be non-linear. The Input-Output relation of a non-linear device can be expressed as

$$V_o = a_0 + a_1 V_{in} + a_2 V_{in}^2 + a_3 V_{in}^3 + a_4 V_{in}^4 + \dots\dots\dots$$

When the input is very small, the higher power terms can be neglected. Hence the output is approximately given by $V_o = a_0 + a_1 V_{in} + a_2 V_{in}^2$

When the output is considered up to square of the input, the device is called a square law device and the square law modulator is as shown in the figure 4

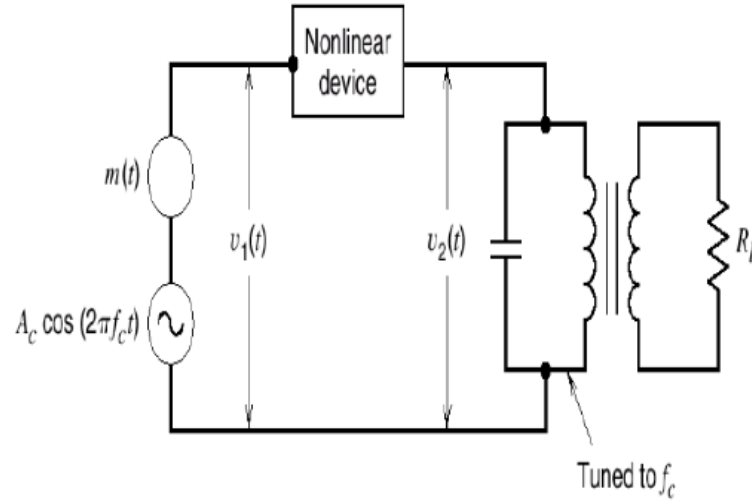


Fig.4. Square Law Modulator

Consider a non-linear device to which a carrier $c(t) = A_c \cos(2\pi f_c t)$ and an information signal $m(t)$ are fed simultaneously as shown in figure 4. The total input to the device at any instant is

$$V_{in} = c(t) + m(t)$$

$$V_{in} = A_c \cos 2\pi f_c t + m(t)$$

As the level of the input is very small, the output can be considered up to square of the input, i.e., $V_o = a_0 + a_1 V_{in} + a_2 V_{in}^2$

$$V_o = a_0 + a_1 [A_c \cos 2\pi f_c t + m(t)] + a_2 [A_c \cos 2\pi f_c t + m(t)]^2$$

$$V_o = a_0 + a_1 A_c \cos 2\pi f_c t + a_1 m(t) + \frac{a_2 A_c^2}{2} (1 + \cos 4\pi f_c t) + a_2 [m(t)]^2 + 2a_2 m(t) A_c \cos 2\pi f_c t$$

$$V_o = a_0 + a_1 A_c \cos 2\pi f_c t + a_1 m(t) + \frac{a_2 A_c^2}{2} \cos 4\pi f_c t + a_2 m^2(t) + 2a_2 m(t) A_c \cos 2\pi f_c t$$

Taking Fourier transform on both sides, we get

$$V_o(f) = (a_0 + \frac{a_2 A_c^2}{2}) \delta(f) + \frac{a_1 A_c}{2} [\delta(f - f_c) + \delta(f + f_c)] + a_1 M(f) +$$

$$\frac{a_2 A_c^2}{4} [\delta(f - 2f_c) + \delta(f + 2f_c)] + a_2 M(f) + a_2 A_c [M(f - f_c) + M(f + f_c)]$$

Therefore the square law device output 0 V consists of the dc component at $f = 0$. The information signal ranging from 0 to W Hz and its second harmonics are signal at f_c and $2f_c$.

Frequency band centered at f_c with a deviation of $\pm W$, Hz.

The required AM signal with a carrier frequency f_c can be separated using a band pass filter at the out put of the square law device. The filter should have a lower cut-off frequency ranging between $2W$ and $(f_c - W)$ and upper cut-off frequency between $(f_c + W)$ and $2f_c$.

Therefore the filter out put is

$$s(t) = a_1 A_c \cos 2\pi f_c t + 2a_2 A_c m(t) \cos 2\pi f_c t$$

$$s(t) = a_1 A_c \left[1 + 2 \frac{a_2}{a_1} m(t) \right] \cos 2\pi f_c t$$

If $m(t) = A_m \cos 2\pi f_m t$, we get

$$s(t) = a_1 A_c \left[1 + 2 \frac{a_2}{a_1} A_m \cos 2\pi f_m t \right] \cos 2\pi f_c t$$

Comparing this with the standard representation of AM signal,

$$s(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t)$$

Therefore modulation index of the output signal is given by

$$m = 2 \frac{a_2}{a_1} A_m$$

The output AM signal is free from distortion and attenuation only when $(f_c - W) > 2W$ or $f_c > 3W$.

Spectrum is as shown below

Switching Modulator

Consider a semiconductor diode used as an ideal switch to which the carrier signal $c(t) = A_c \cos(2\pi f_c t)$ and information signal $m(t)$ are applied simultaneously as shown figure

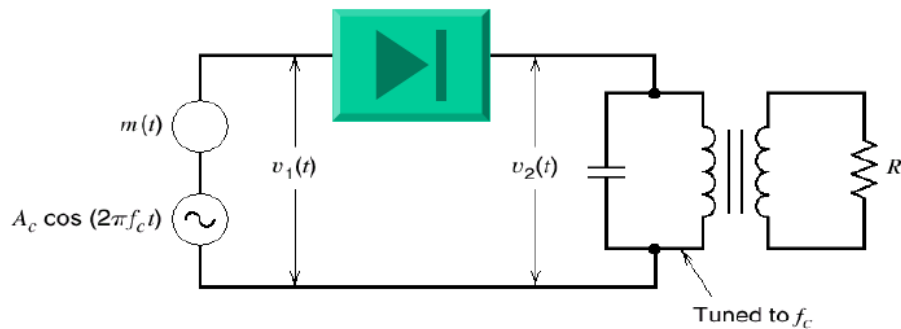


Fig.5. Switching Modulator

The total input for the diode at any instant is given by

$$v_1 = c(t) + m(t)$$

$$v_1 = A_c \cos 2\pi f_c t + m(t)$$

When the peak amplitude of $c(t)$ is maintained more than that of information signal, the operation is assumed to be dependent on only $c(t)$ irrespective of $m(t)$.

When $c(t)$ is positive, $v_2=v_1$ since the diode is forward biased. Similarly, when $c(t)$ is negative, $v_2=0$ since diode is reverse biased. Based upon above operation, switching response of the diode is periodic rectangular wave with an amplitude unity and is given by

$$p(t) = \frac{1}{2} + \frac{1}{\pi} \sum_{n=-\infty}^{\infty} \frac{(-1)^{n-1}}{2n-1} \cos(2\pi f_c t (2n-1))$$

$$p(t) = \frac{1}{2} + \frac{2}{\pi} \cos(2\pi f_c t) - \frac{2}{3\pi} \cos(6\pi f_c t) + -$$

$n=0,1$ $n=-1,+2$

Therefore the diode response V_o is a product of switching response $p(t)$ and input v_1 .

$$v_2 = v_1 * p(t)$$

$$V_2 = [A_c \cos 2\pi f_c t + m(t)] \left[\frac{1}{2} + \frac{2}{\pi} \cos 2\pi f_c t - \frac{2}{3\pi} \cos 6\pi f_c t + - + - \right]$$

Applying the Fourier Transform, we get

$$\begin{aligned} V_2(f) = & \frac{A_c}{4} [\delta(f - f_c) + \delta(f + f_c)] + \frac{M(f)}{2} + \frac{A_c}{\pi} \delta(f) \\ & + \frac{A_c}{2\pi} [\delta(f - 2f_c) + \delta(f + 2f_c)] + \frac{1}{\pi} [M(f - f_c) + M(f + f_c)] \\ & - \frac{A_c}{6\pi} [\delta(f - 4f_c) + \delta(f + 4f_c)] - \frac{A_c}{3\pi} [\delta(f - 2f_c) + \delta(f + 2f_c)] \\ & - \frac{1}{3\pi} [M(f - 3f_c) + M(f + f_c)] \end{aligned}$$

The diode output v_2 consists of

a dc component at $f=0$.

Information signal ranging from 0 to w Hz and infinite number of frequency bands centered at $f, 2f_c, 3f_c, 4f_c, \dots$

The required AM signal centred at f_c can be separated using band pass filter. The lower cut off-frequency for the band pass filter should be between w and $f_c - w$ and the upper cut-off frequency between $f_c + w$ and $2f_c$. The filter output is given by the equation

$$S(t) = \frac{A_c}{2} \left[1 + \frac{4}{\pi} \frac{m(t)}{A_c} \right] \cos 2\pi f_c t$$

For a single tone information, let $m(t) = A_m \cos(2\pi f_m t)$

$$S(t) = \frac{A_c}{2} \left[1 + \frac{4}{\pi} \frac{A_m}{A_c} \cos 2\pi f_m t \right] \cos 2\pi f_c t$$

Therefore modulation index, $m = \frac{4}{\pi} \frac{A_m}{A_c}$

The output AM signal is free from distortions and attenuations only when $f_c - w > w$ or $f_c > 2w$.

Detection of AM waves

Demodulation is the process of recovering the information signal (base band) from the incoming modulated signal at the receiver. There are two methods, they are Square law Detector and Envelope Detector

Square Law Detector

Consider a non-linear device to which the AM signal $s(t)$ is applied. When the level of $s(t)$ is very small, output can be considered upto square of the input.

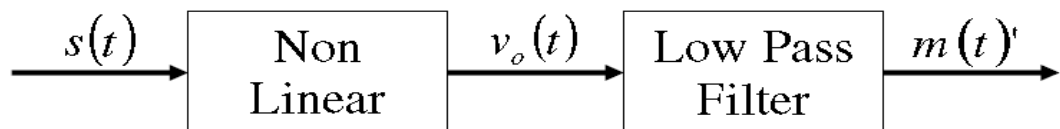


Figure: Demodulation of AM using square law device

$$\text{Therefore } V_o = a_o + a_1 V_m + a_2 V_m^2$$

If $m(t)$ is the information signal (0-wHz) and $c(t) = A_c \cos(2\pi f_c t)$ is the carrier, input AM signal to the non-linear device is given by

$$s(t) = A_c [1 + k_a m(t)] \cos(2\pi f_c t)$$

Superheterodyne Receiver

The received RF-signals must be transformed into a video signal to get the wanted information from the echoes. This transformation is made by a superheterodyne receiver. The main components of the typical superheterodyne receiver are shown on the following picture:

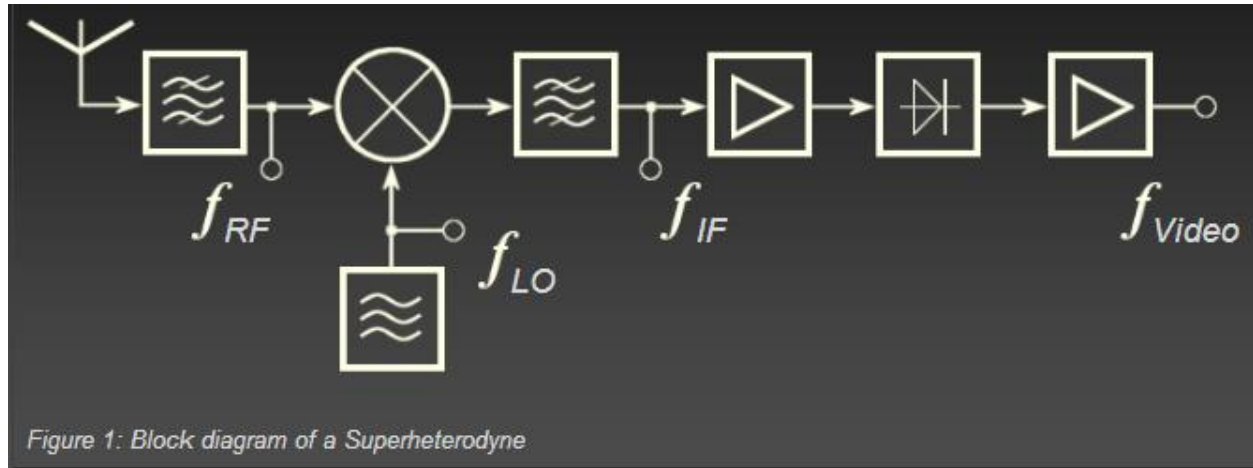


Figure 1: Block diagram of a Superheterodyne

The superheterodyne receiver changes the rf frequency into an easier to process lower IF- frequency. This IF- frequency will be amplified and demodulated to get a video signal.

The Figure shows a block diagram of a typical superheterodyne receiver. The RF-carrier comes in from the antenna and is applied to a filter. The output of the filter are only the frequencies of the desired frequency-band. These frequencies are applied to the mixer stage. The mixer also receives an input from the local oscillator. These two signals are beat together to obtain the IF through the process of heterodyning. There is a fixed difference in frequency between the local oscillator and the rf-signal at all times by tuning the local oscillator. This difference in frequency is the IF. this fixed difference is ganged tuning ensures a constant IF over the frequency range of the receiver. The IF-carrier is applied to the IF-amplifier. The amplified IF is then sent to the detector. The output of the detector is the video component of the input signal.

Image-frequency Filter

A low-noise RF amplifier stage ahead of the converter stage provides enough selectivity to reduce the image-frequency response by rejecting these unwanted signals and adds to the sensitivity of the receiver. The borders of the bandwidth of this amplifier are chosen to eliminate the image frequencies.

Many older radar receivers do not use a low-noise pre-amplifier (RF stage) as the receiver front end; they simply send the echo signal directly to a crystal mixer stage. This has any disadvantages. It is possible for these receivers to receive two different stations at the same point of the dial.

Mixer Stage

The mixer stage is used to increase the received frequency to an intermediate frequency. The mixer also receives an input from the local oscillator. These two signals are beat together to obtain the IF through the process of heterodyning.

$$f_{IF} = f_{rx} - f_{local}$$

oscillator

$$f_{IF} = f_{local}$$

$$\text{oscillator} - f_{rx}$$

There aren't any components which can distinguish a negative frequency of a positive frequency. therefore we can measure the magnitude of the frequency only: $f_{IF} = |f_{local} \text{ oscillator} - f_{rx}|$

The result is a second reception frequency as a „mirror image“ around the intermediate frequency.

Assuming an intermediate frequency of 60 MHz, the local oscillator will track at a frequency of 60 MHz higher than the incoming signal. For example, suppose the receiver is tuned to pick up a signal on a frequency of 1030 MHz. The local oscillator will be operating at a frequency of 1090 MHz. The received and local oscillator signals are mixed, or heterodyned, in the converter stage and one of the frequencies resulting from this mixing action

is the difference between the two signals, or 60 MHz, the IF frequency. This IF frequency is then amplified in the IF stages and sent on to the detector and audio stages.

Any signal at a frequency of 60 MHz that appears on the plate of the converter circuit will be accepted by the IF amplifier and passed on.

So on a receiver with no RF amplifier, the input to the converter is rather broadly tuned and some signals other than the desired signal will get through to the input jack of the converter stage. Normally these other signals will mix with the local oscillator signal and produce frequencies that are outside the bandpass of the 60 MHz IF amplifier and will be rejected. However, if there is a station operating on a frequency of 1150 MHz, and this signal passes through the rather broad tuned input circuit and appears on the input jack of the converter stage, it too will mix with the local oscillator and produce a frequency of 60 MHz ($1150 - 1090 = 60$). This signal will also be accepted by the IF amplifier stage and passed on, thus both signals will be indicated on the screen. This is known as image-frequency interference.

IF Filter

This filter must filter the desired intermediate frequency out from the mixture frequencies arisen in the mixer stage. It is designed as one or more bandpasses.

Normally, the bandpass is as narrow as possible without affecting the actual signal energy. When a selection of pulse widths is available, such as short and long pulses, the bandpass must be able to match the bandwidth of the two different signals.

IF- Amplifier

The IF amplifier has the capability to vary both the bandpass and the gain of a receiver. After conversion to the intermediate frequency, the signal is amplified in several IF- amplifier stages. Most of the gain of the receiver is developed in the IF amplifier stages. The overall bandwidth of the receiver is often determined by the bandwidth of the IF stages. Gain must be variable to provide a constant voltage output for input signals of different amplitudes.

Detector

Scan from a screen of an oscilloscope.

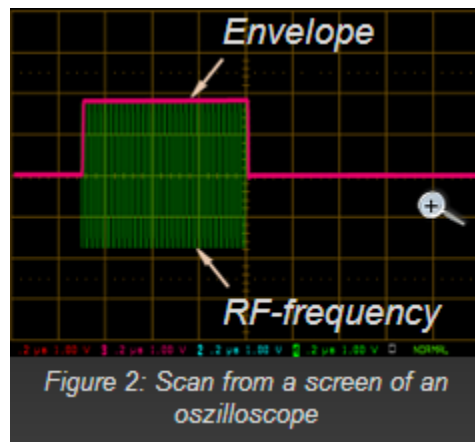


Figure 2: Scan from a screen of an oscilloscope

The detector in a microwave receiver serves to convert the IF pulses into video pulses.

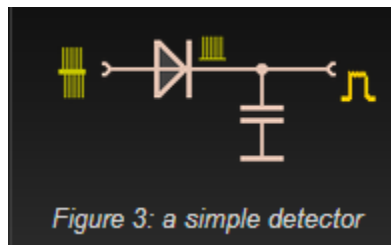


Figure 3: a simple detector

The simplest form of detector is the diode detector. It detects the pulse envelope:

The condenser has got the function of a lowpass and blocks the IF-frequency.

In addition to the shown Amplitude Modulation there are possible other types of modulation too.

Video Amplifier

The video amplifier receives pulses from the detector and amplifies these pulses for application to the indicating device. A video amplifier is fundamentally an RC coupled amplifier that uses high-gain transistors. However, a video amplifier must be capable of a relatively wide Frequency response. The output stage of the receiver is normally an emitter follower.

The low-impedance output of the emitter follower matches the impedance of the cable. The video pulses are coupled through the cable to the indicator for video display on the crt.

Local Oscillator

The local oscillator excite a frequency for mixing with the incoming signal to get the intermediate frequency.

Most radar receivers use megahertz intermediate frequency (IF) with a value between 30 and 75 megahertz. The IF is produced by mixing a local oscillator signal with the incoming signal. The local oscillator is, therefore, essential to efficient operation and must be both tunable and very stable. For example, if the local oscillator frequency is 3,000 megahertz, a frequency change of 0.1 percent will produce a frequency shift of 3 megahertz. This is equal to the bandwidth of most receivers and would greatly decrease receiver gain.

The power output requirement for most local oscillators is small (20 to 50 milliwatts) because most receivers use crystal mixers that require very little power.

The local oscillator output frequency must be tunable over a range of several megahertz in the 4,000-megahertz region. The local oscillator must compensate for any changes in the transmitted frequency and maintain a constant 30 or 75 megahertz difference between the oscillator and the transmitter frequency. A local oscillator that can be tuned by varying the applied voltage is most desirable.

The exiting frequency is either higher or lower than the incoming frequency. An RF amplifier stage ahead of the converter stage provides enough selectivity to reduce the image-frequency response by rejecting these unwanted signals and adds to the sensitivity of the receiver.

OVERVIEW OF OPTICAL FIBER COMMUNICATION: INTRODUCTION

Introduction

Fiber-optic communication is a method of transmitting information from one place to another by sending pulses of light through an optical fiber. The light forms an electromagnetic carrier wave that is modulated to carry information.^[1] Fiber is preferred over electrical cabling when high bandwidth, long distance, or immunity to electromagnetic interference are required. This type of communication can transmit voice, video, and telemetry through local area networks, computer networks, or across long distances.

Optical fiber is used by many telecommunications companies to transmit telephone signals, Internet communication, and cable television signals. Researchers at Bell Labs have reached internet speeds of over 100 peta bit ×kilometer per second using fiber-optic communication.

The process of communicating using fiber-optics involves the following basic steps:

1. creating the optical signal involving the use of a transmitter, usually from an electrical signal
2. relaying the signal along the fiber, ensuring that the signal does not become too distorted or weak
3. receiving the optical signal
4. converting it into an electrical signal

Historical Development

First developed in the 1970s, fiber-optics have revolutionized the telecommunications industry and have played a major role in the advent of the Information Age. Because of its advantages over electrical transmission, optical fibers have largely replaced copper wire communications in core networks in the developed world.

In 1880 Alexander Graham Bell and his assistant Charles Sumner Tainter created a very early precursor to fiber-optic communications, the Photophone, at Bell's newly established Volta Laboratory in Washington, D.C. Bell considered it his most important invention. The device allowed for the transmission of sound on a beam of light. On June 3, 1880, Bell conducted the world's first wireless telephone transmission between two buildings, some 213 meters apart.^{[4][5]} Due to its use of an atmospheric transmission medium, the Photophone would not prove practical until advances in laser and optical fiber technologies permitted the secure transport of light. The Photophone's first practical use came in military communication systems many decades later.

In 1954 Harold Hopkins and Narinder Singh Kapany showed that rolled fiber glass allowed light to be transmitted. Initially it was considered that the light can traverse in only straight medium. Jun-ichi Nishizawa, a Japanese scientist at Tohoku University, proposed the use of optical fibers for communications in 1963. Nishizawa invented the PIN diode and the static induction transistor, both of which contributed to the development of optical fiber communications.

In 1966 Charles K. Kao and George Hockham at STC Laboratories (STL) showed that the losses of 1,000 dB/km in existing glass (compared to 5–10 dB/km in coaxial cable) were due to contaminants which could potentially be removed.

Optical fiber was successfully developed in 1970 by Corning Glass Works, with attenuation low enough for communication purposes (about 20 dB/km) and at the same time GaAs semiconductor lasers were developed that were compact and therefore suitable for transmitting light through fiber optic cables for long distances.

In 1973, Optelecom, Inc., co-founded by the inventor of the laser, Gordon Gould, received a contract from APA for the first optical communication systems. Developed for Army Missile Command in Huntsville, Alabama, it was a laser on the ground and a spout of optical fiber played out by missile to transmit a modulated signal over five kilometers.

After a period of research starting from 1975, the first commercial fiber-optic communications system was developed which operated at a wavelength around $0.8\text{ }\mu\text{m}$ and used GaAs semiconductor lasers. This first-generation system operated at a bit rate of 45 Mbit/s with repeater spacing of up to 10 km. Soon on 22 April 1977, General Telephone and Electronics sent the first live telephone traffic through fiber optics at a 6 Mbit/s throughput in Long Beach, California.

In October 1973, Corning Glass signed a development contract with CSELT and Pirelli aimed to test fiber optics in an urban environment: in September 1977, the second cable in this test series, named COS-2, was experimentally deployed in two lines (9 km) in Turin, for the first time in a big city, at a speed of 140 Mbit/s.

The second generation of fiber-optic communication was developed for commercial use in the early 1980s, operated at $1.3\text{ }\mu\text{m}$ and used InGaAsP semiconductor lasers. These early systems were initially limited by multi mode fiber dispersion, and in 1981 the single-mode fiber was revealed to greatly improve system performance, however practical connectors capable of working with single mode fiber proved difficult to develop. Canadian service provider SaskTel had completed construction of what was then the world's longest commercial fiber optic network, which covered 3,268 km (2,031 mi) and linked 52 communities.^[11] By 1987, these systems were operating at bit rates of up to 1.7 Gb/s with repeater spacing up to 50 km (31 mi).

The first transatlantic telephone cable to use optical fiber was TAT-8, based on Desurvire optimised laser amplification technology. It went into operation in 1988.

Third-generation fiber-optic systems operated at $1.55\text{ }\mu\text{m}$ and had losses of about 0.2 dB/km. This development was spurred by the discovery of Indium gallium arsenide and the development of the Indium Gallium Arsenide photodiode by Pearsall. Engineers overcame earlier difficulties with pulse-spreading at that wavelength using conventional InGaAsP semiconductor lasers. Scientists overcame this difficulty by using dispersion-shifted fibers designed to have minimal dispersion at $1.55\text{ }\mu\text{m}$ or by limiting the laser spectrum to a single longitudinal mode.

These developments eventually allowed third-generation systems to operate commercially at 2.5 Gbit/s with repeater spacing in excess of 100 km (62 mi).

The fourth generation of fiber-optic communication systems used optical amplification to reduce the need for repeaters and wavelength-division multiplexing to increase data capacity. These two improvements caused a revolution that resulted in the doubling of system capacity every six months starting in 1992 until a bit rate of 10 Tb/s was reached by 2001. In 2006 a bit-rate of 14 Tbit/s was reached over a single 160 km (99 mi) line using optical amplifiers.

The focus of development for the fifth generation of fiber-optic communications is on extending the wavelength range over which a WDM system can operate. The conventional wavelength window, known as the C band, covers the wavelength range 1.53–1.57 μm , and *dry fiber* has a low-loss window promising an extension of that range to 1.30–1.65 μm . Other developments include the concept of "optical solutions", pulses that preserve their shape by counteracting the effects of dispersion with the nonlinear effects of the fiber by using pulses of a specific shape.

In the late 1990s through 2000, industry promoters, and research companies such as KMI, and RHK predicted massive increases in demand for communications bandwidth due to increased use of the Internet, and commercialization of various bandwidth-intensive consumer services, such as video on demand. Internet protocol data traffic was increasing exponentially, at a faster rate than integrated circuit complexity had increased under Moore's Law. From the bust of the dot-com bubble through 2006, however, the main trend in the industry has been consolidation of firms and offshoring of manufacturing to reduce costs. Companies such as Verizon and AT&T have taken advantage of fiber-optic communications to deliver a variety of high-throughput data and broadband services to consumers' homes.

Advantages of Fiber Optic Transmission

Optical fibers have largely replaced copper wire communications in core networks in the developed world, because of its advantages over electrical transmission. Here are the main advantages of fiber optic transmission.

Extremely High Bandwidth: No other cable-based data transmission medium offers the bandwidth that fiber does. The volume of data that fiber optic cables transmit per unit time is far greater than copper cables.

Longer Distance: in fiber optic transmission, optical cables are capable of providing low power loss, which enables signals can be transmitted to a longer distance than copper cables.

Resistance to Electromagnetic Interference: in practical cable deployment, it's inevitable to meet environments like power substations, heating, ventilating and other industrial sources of interference. However, fiber has a very low rate of bit error (10⁻¹³), as a result of fiber being so resistant to electromagnetic interference. Fiber optic transmission is virtually noise free.

Low Security Risk: the growth of the fiber optic communication market is mainly driven by increasing awareness about data security concerns and use of the alternative raw material. Data or signals are transmitted via light in fiber optic transmission. Therefore there is no way to detect the data being transmitted by "listening in" to the electromagnetic energy "leaking" through the cable, which ensures the absolute security of information.

Small Size: fiber optic cable has a very small diameter. For instance, the cable diameter of a single OM3 multimode fiber is about 2mm, which is smaller than that of coaxial copper cable. Small size saves more space in fiber optic transmission.

Light Weight: fiber optic cables are made of glass or plastic, and they are thinner than copper cables. These make them lighter and easy to install.

Easy to Accommodate Increasing Bandwidth: with the use of fiber optic cable, new equipment can be added to existing cable infrastructure. Because optical cable can provide vastly expanded capacity over the originally laid cable. And WDM (wavelength division multiplexing) technology, including CWDM and DWDM, enables fiber cables the ability to accommodate more bandwidth.

Disadvantages of Fiber Optic Transmission

Though fiber optic transmission brings lots of convenience, its disadvantages also cannot be ignored. **Fragility:** usually optical fiber cables are made of glass, which lends to they are more fragile than electrical wires. In addition, glass can be affected by various chemicals including hydrogen gas (a problem in underwater cables), making them need more cares when deployed under ground.

Difficult to Install: it's not easy to splice fiber optic cable. And if you bend them too much, they will break. And fiber cable is highly susceptible to becoming cut or damaged during installation or construction activities. All these make it difficult to install.

Attenuation & Dispersion: as transmission distance getting longer, light will be attenuated and dispersed, which requires extra optical components like EDFA to be added.

Cost Is Higher Than Copper Cable: despite the fact that fiber optic installation costs are dropping by as much as 60% a year, installing fiber optic cabling is still relatively higher than copper cables. Because copper cable installation does not need extra care like fiber cables. However, optical fiber is still moving into the local loop, and through technologies such as FTTx (fiber to the home, premises, etc.) and PONs (passive optical networks), enabling subscriber and end user broadband access.

Special Equipment Is Often Required: to ensure the quality of fiber optic transmission, some special equipment is needed. For example, equipment such as OTDR (optical time-domain reflectometry) is required and expensive, specialized optical test equipment such as optical probes and power meter are needed at most fiber endpoints to properly provide testing of optical fiber.

Applications of Optical Fiber Communications

Fiber optic cables find many uses in a wide variety of industries and applications. Some uses of fiber optic cables include:

- **Medical**

Used as light guides, imaging tools and also as lasers for surgeries

- **Defense/Government**

Used as hydrophones for seismic waves and SONAR , as wiring in aircraft, submarines and other vehicles and also for field networking

- **Data Storage**

Used for data transmission

- **Telecommunications**

Fiber is laid and used for transmitting and receiving purposes

- **Networking**

Used to connect users and servers in a variety of network settings and help increase the speed and accuracy of data transmission

- **Industrial/Commercial**

Used for imaging in hard to reach areas, as wiring where EMI is an issue, as sensory devices to make temperature, pressure and other measurements, and as wiring in automobiles and in industrial settings

- **Broadcast/CATV**

Broadcast/cable companies are using fiber optic cables for wiring CATV, HDTV, internet, video on-demand and other applications

Fiber optic cables are used for lighting and imaging and as sensors to measure and monitor a vast array of variables. Fiber optic cables are also used in research and development and testing across all the above mentioned industries

The optical fibers have many applications. Some of them are as follows –

- Used in telephone systems
- Used in sub-marine cable networks
- Used in data link for computer networks, CATV Systems
- Used in CCTV surveillance cameras
- Used for connecting fire, police, and other emergency services.
- Used in hospitals, schools, and traffic management systems.
- They have many industrial uses and also used for in heavy duty constructions.

Block Diagram of Optical Fiber Communication System

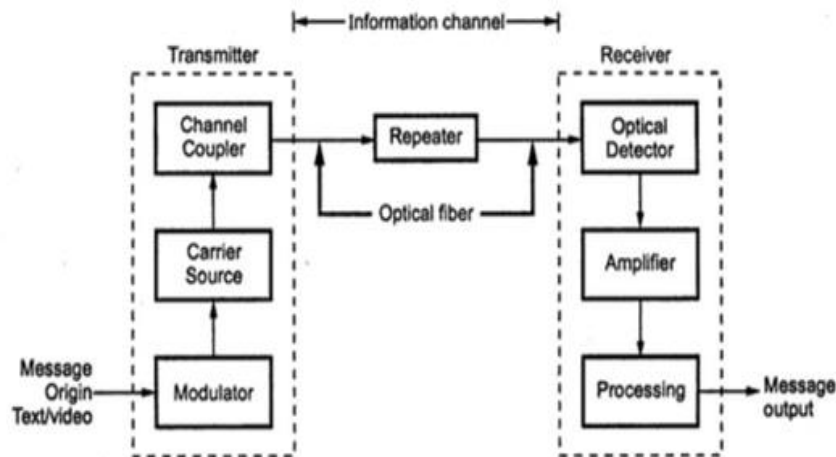


Fig 1: Block Diagram of Optical Fiber Communication System

Message origin:

Generally message origin is from a transducer that converts a non-electrical message into an electrical signal. Common examples include microphones for converting sound waves into currents and video (TV) cameras for converting images into current. For data transfer between computers, the message is already in electrical form.

Modulator:

The modulator has two main functions.

- 1) It converts the electrical message into proper format.
- 2) It impresses this signal onto the wave generated by the carrier source.

Two distinct categories of modulation are used i.e. analog modulation and digital modulation.

Carrier source:

. Carrier source generates the wave on which the information is transmitted. This wave is called the carrier. For fiber optic system, a laser diode (LD) or a light emitting diode (LED) is used. They can be called as optic oscillators, they provide stable, single frequency waves with sufficient power for long distance propagation.

Channel coupler:

. Coupler feeds the power into information channel. For an atmospheric optic system, the channel coupler is a lens used for collimating the light emitted by the source and directing this light towards the receiver. The coupler must efficiently transfer the modulated light beam from the source to the optic fiber. The channel coupler design is an important part of fiber system because of possibility of high losses.

Information channel:

. The information channel is the path between the transmitter and receiver. In fiber optic communications, a glass or plastic fiber is the channel. Desirable characteristics of the information channel include low attenuation and large light acceptance cone angle. Optical amplifiers boost the power levels of weak signals. Amplifiers are needed in very long links to provide sufficient power to the receiver. Repeaters can be used only for digital systems. They convert weak and distorted optical signals to electrical ones and then regenerate the original digital pulse trains for further transmission.

. Another important property of the information channel is the propagation time of the waves travelling along it. A signal propagating along a fiber normally contains a range of fiber optic frequencies and divides its power along several ray paths. This results in a distortion of the propagation signal. In a digital system, this distortion appears as a spreading and deforming of the pulses. The spreading is so great that adjacent pulses begin to overlap and become unrecognizable as separate bits of information.

Optical detector:

. The information begin transmitted is detected by detector. In the fiber system the optic wave is converted into an electric current by a photodetector. The current developed by the detector is proportional to the power in the incident optic wave. Detector output current contains the transmitted information. This detector output is then filtered to remove the constant bias and then amplified.

. The important properties of photodetectors are small size, economy, long life, low power consumption, high sensitivity to optic signals and fast response to quick variations in the optic power.

. Signal processing includes filtering, amplification. Proper filtering maximizes the ratio of signal to unwanted power. For a digital system decision circuit is an additional block. The bit error rate (BER) should be very small for quality communications.

Signal processing:

. Signal processing includes filtering, amplification. Proper filtering maximizes the ratio of signal to unwanted power. For a digital system decision circuit is an additional block. The bit error rate (BER) should be very small for quality communications.

Message output:

. The electrical form of the message emerging from the signal processor is transformed into a sound wave or visual image. Sometimes these signals are directly usable when computers or other machines are connected through a fiber system.

Electromagnetic Spectrum

The radio waves and light are electromagnetic waves. The rate at which they alternate in polarity is called their frequency (f) measured in hertz (Hz). The speed of electromagnetic wave (c) in free space is approximately 3×10^8 m/sec. The distance travelled during each cycle is called as wavelength (λ)

In fiber optics, it is more convenient to use the wavelength of light instead of the frequency with light frequencies; wavelength is often stated in microns or nanometers.

1 micron (μ) = 1

Micrometre (1×10^{-6}) 1 nano (n) = 10^{-9} meter

Fiber optics uses visible and infrared light. Infrared light covers a fairly wide range of wavelengths and is generally used for all fiber optic communications. Visible light is normally used for very short range transmission using a plastic fiber.

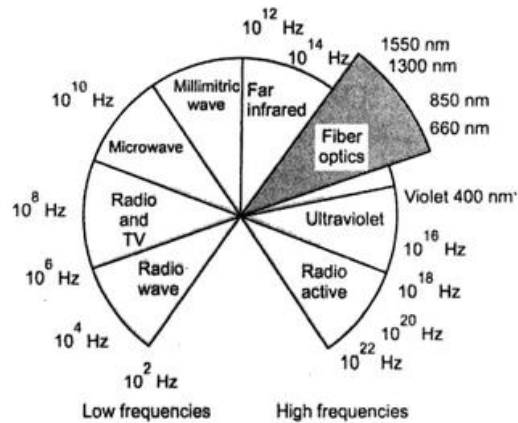
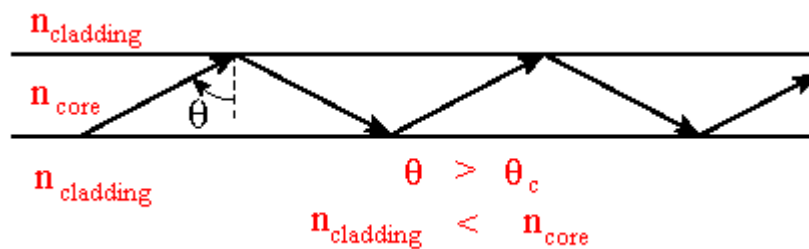


Fig 2: Electromagnetic Spectrum

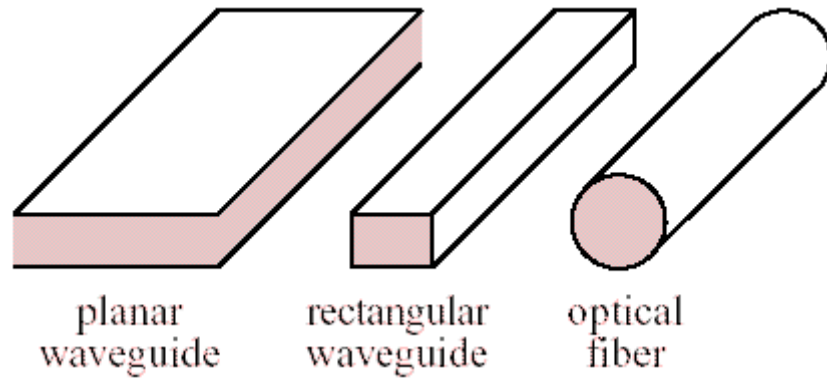
Optical Fiber Waveguides

In free space light travels at its maximum possible speed i.e. 3×10^8 m/s or 186 x 10³ miles/sec. When light travels through a material it exhibits certain behavior explained by laws of reflection, refraction.

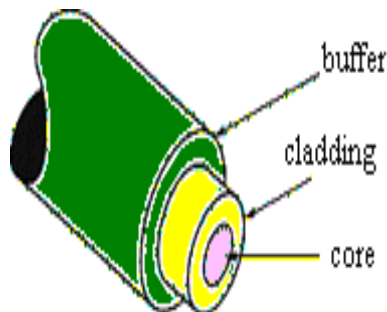
An optical wave guide is a structure that "guides" a light wave by constraining it to travel along a certain desired path. If the transverse dimensions of the guide are much larger than the wavelength of the guided light, then we can explain how the optical waveguide works using geometrical optics and total internal reflection.



A wave guide traps light by surrounding a guiding region, called the core, made from a material with index of refraction n_{core} , with a material called the cladding, made from a material with index of refraction $n_{\text{cladding}} < n_{\text{core}}$. Light entering is trapped as long as $\sin\theta > n_{\text{cladding}}/n_{\text{core}}$.

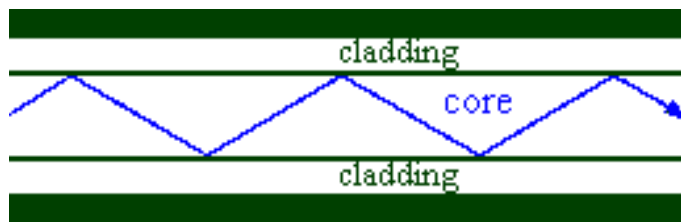


Light can be guided by planar or rectangular wave guides, or by optical fibers. An optical fiber consists of three concentric elements, the core, the cladding and the outer coating, often called the buffer. The core is usually made of glass or plastic. The core is the light-carrying portion of the fiber. The cladding surrounds the core. The cladding is made of a material with a slightly lower index of refraction than the core. This difference in the indices causes total internal reflection to occur at the core-cladding boundary along the length of the fiber. Light is transmitted down the fiber and does not escape through the sides of the fiber.

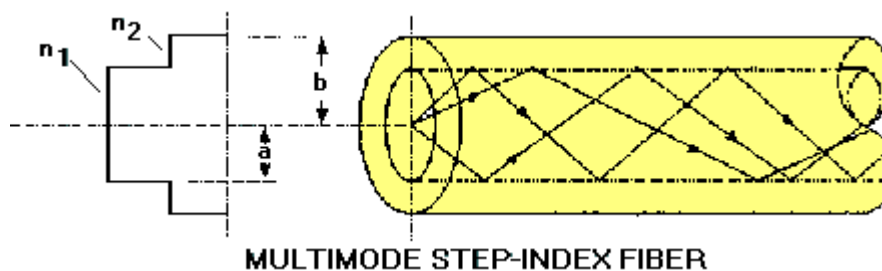


- Fiber Optic Core:
 - the inner light-carrying member with a high index of refraction.
- Cladding:
 - the middle layer, which serves to confine the light to the core. It has a lower index of refraction.

- Buffer:
 - the outer layer, which serves as a "shock absorber" to protect the core and cladding from damage. The coating usually comprises one or more coats of a plastic material to protect the fiber from the physical environment. Sometimes metallic sheaths are added to the coating for further physical protection.



-
- Light injected into the fiber optic core and striking the core-to-cladding interface at an angle greater than the critical angle is reflected back into the core. Since the angles of incidence and reflection are equal, the light ray continues to zigzag down the length of the fiber. The light is trapped within the core. Light striking the interface at less than the critical angle passes into the cladding and is lost.



- Fibers for which the refractive index of the core is a constant and the index changes abruptly at the core-cladding interface are called step-index fibers. Step-index fibers are available with core diameters of 100 μm to 1000 μm . They are well suited to applications requiring high-power densities, such as delivering laser power for medical and industrial applications.

Lecture-6
Satellite Communication

AIM & OBJECTIVE

- ❖ To understand the basics of satellite orbits.
- ❖ To analyze the geo stationary and non geo stationary orbits.
- ❖ To acquire the knowledge about launching procedures.

PRE-TEST MCQ

1. A television (TV) transmission is an example of which type of transmission?
a) Simplex
b) Half duplex
c) Full duplex
d) None of the above

2. What is application of satellite systems?
a) Weather forecasting
b) Terrestrial communication
c) point to point communication
d) None of the above

3. The down link frequency in the C band transponder is
(a) 6 GHz
(b) 4 GHz
(c) 14 GHz
(d) 11 GHz

THEORY

1.1 Introduction to satellite communication

Satellites are specifically made for telecommunication purpose. They are used for mobile applications such as communication to ships, vehicles, planes, hand-held terminals and for TV and radio broadcasting.

They are responsible for providing these services to an assigned region (area) on the earth. The power and bandwidth of these satellites depend upon the preferred size of the footprint, complexity of the traffic control protocol schemes and the cost of ground stations.

A satellite works most efficiently when the transmissions are focused with a desired area.

When the area is focused, then the emissions don't go outside that designated area and thus minimizing the interference to the other systems. This leads more efficient spectrum usage.

Satellites antenna patterns play an important role and must be designed to best cover the designated geographical area (which is generally irregular in shape).

Satellites should be designed by keeping in mind its usability for short and long term effects throughout its life time.

The earth station should be in a position to control the satellite if it drifts from its orbit it is subjected to any kind of drag from the external forces.

Applications of Satellites:

- ❖ Weather Forecasting
- ❖ Radio and TV Broadcast
- ❖ Military Satellites
- ❖ Navigation Satellites
- ❖ Global Telephone
- ❖ Connecting Remote Area
- ❖ Global Mobile Communication

1.2 Kepler's laws

1.2.1 Kepler's law Introduction

Satellites (spacecraft) orbiting the earth follow the same laws that govern the motion of the planets around the sun.

Kepler's laws apply quite generally to any two bodies in space which interact through gravitation. The more massive of the two bodies is referred to as the *primary*, the other, the *secondary* or *satellite*.

1.2.2 Kepler's First Law

Kepler's first law states that the path followed by a satellite around the primary will be an ellipse. An ellipse has two focal points shown as F_1 and F_2 in Fig. 2.1. The center of mass of the two-body system, termed the *bary center*, is always center of the foci.

The semi major axis of the ellipse is denoted by a , and the semi minor axis, by b . The eccentricity e is given by

$$e = \frac{\sqrt{a^2 - b^2}}{a}$$

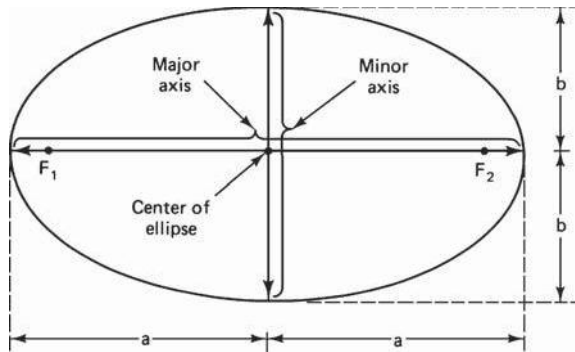


Figure 1.1 The foci F_1 and F_2 , the semi major axis a , and the semi minor axis b of an ellipse.

1.2.3 Kepler's Second Law

Kepler's second law states that, for equal time intervals, a satellite will sweep out equal areas in its orbital plane, focused at the barycenter. Referring to Fig. 2.2, assuming the satellite travels distances S_1 and S_2 meters in 1 s, then the areas A_1 and A_2 will be equal. The average velocity in each case is S_1 and S_2 m/s, and because of the equal area law, it follows that the velocity at S_2 is less than that at S_1 .

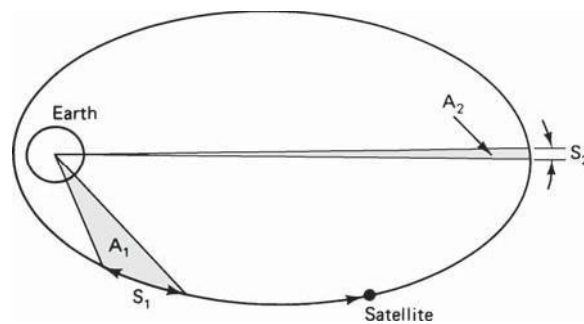


Figure 1.2 Kepler's second law. The areas A_1 and A_2 swept out in unit time are equal.

1.2.4 Kepler's Third Law

Kepler's third law states that the square of the periodic time of orbit is proportional to the cube of the mean distance between the two bodies. The mean distance is equal to the semi major axis a .

For the artificial satellites orbiting the earth, Kepler's third law can be written in the form

$$a^3 = \frac{\mu}{n^2}$$

Where n is the mean motion of the satellite in radians per second and μ is the earth's geocentric gravitational constant $\mu = 3.986005 \times 10^{14} \text{ m}^3/\text{s}^2$

1.3. Newton's law:

1.3.1 Newton's first law

An object at rest will remain at rest unless acted on by an unbalanced force. An object in motion continues in motion with the same speed and in the same direction unless acted upon by an unbalanced force. This law is often called "**the law of inertia**".

1.3.2 Newton's second law

Acceleration is produced when a force acts on a mass. The greater the mass (of the object being accelerated) the greater the amount of force needed (to accelerate the object).

1.3.3 Newton's first law

For every action there is an equal and opposite re-action. This means that for every force there is a reaction force that is equal in size, but opposite in direction. That is to say that whenever an object pushes another object it gets pushed back in the opposite direction equally hard.

1.4. orbital parameters

Apogee: A point for a satellite farthest from the Earth. It is denoted as **ha**.

Perigee: A point for a satellite closest from the Earth. It is denoted as **hp**.

Line of Apsides: Line joining perigee and apogee through centre of the Earth. It is the major axis of the orbit. One-half of this line's length is the semi-major axis equivalent to satellite's mean distance from the Earth.

Ascending Node: The point where the orbit crosses the equatorial plane going from north to south.

Descending Node: The point where the orbit crosses the equatorial plane going from south to north.

Inclination: the angle between the orbital plane and the Earth's equatorial plane. Its measured at the ascending node from the equator to the orbit, going from East to North. Also, this angle is commonly denoted as **i**.

Line of Nodes: the line joining the ascending and descending nodes through the centre of Earth.

Prograde Orbit: an orbit in which satellite moves in the same direction as the Earth's rotation. Its inclination is always between 00 to 900. Many satellites follow this path as Earth's velocity makes it easier to launch these satellites.

Retrograde Orbit: an orbit in which satellite moves in the same direction counter to the Earth's rotation.

Argument of Perigee: An angle from the point of perigee measure in the orbital plane at the Earth's centre, in the direction of the satellite motion.

Right ascension of ascending node: The definition of an orbit in space, the position of ascending node is specified. But as the Earth spins, the longitude of ascending node changes and cannot be used for reference. Thus for practical determination of an orbit, the longitude and time of crossing the ascending node is used. For absolute measurement, a fixed reference point in space is required.

It could also be defined as “*right ascension of the ascending node; right ascension is the angular position measured eastward along the celestial equator from the vernal equinox vector to the hour circle of the object*”.

Mean anomaly: It gives the average value to the angular position of the satellite with reference to the perigee.

True anomaly: It is the angle from perigee to the satellite's position, the angle from point of measure at the Earth's centre.

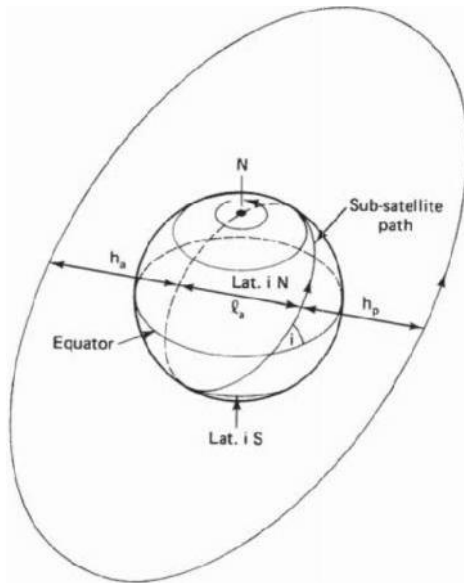


Figure1.2 Apogee height h_a , perigee height h_p , and inclination i . LQ is the line of apsides.

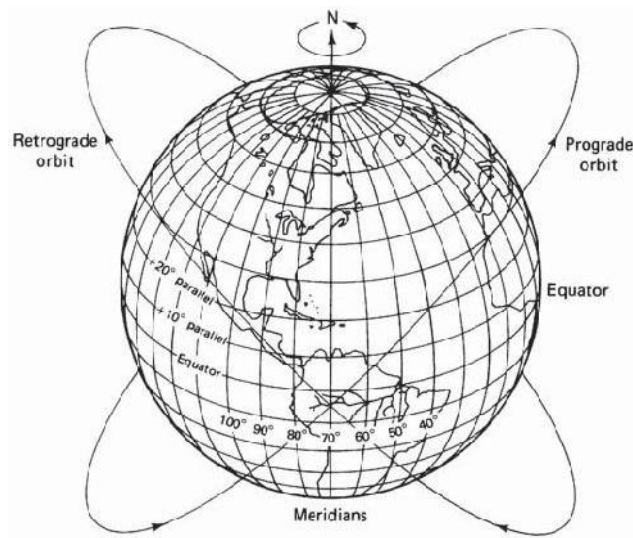


Figure 1.3(a) Prograde and retrograde orbits.

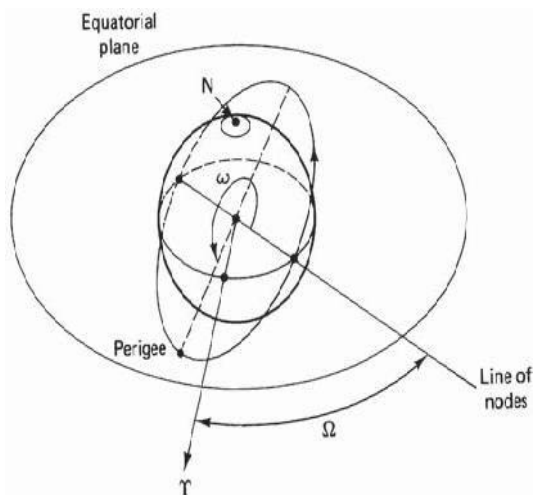


Figure.1.4 The argument of perigee ω & right ascension of the ascending node Ω .

MOBILE COMMUNICATION

Communication is one of the integral parts of science that has always been a focus point for exchanging information among parties at locations physically apart. After its discovery, telephones have replaced the telegrams and letters. Similarly, the term 'mobile' has completely revolutionized the communication by opening up innovative applications that are limited to one's imagination. Today, mobile communication has become the backbone of the society. All the mobile system technologies have improved the way of living. Its main plus point is that it has privileged a common mass of society. In this chapter, the evolution as well as the fundamental techniques of the mobile communication is discussed. The first wireline telephone system was introduced in the year 1877. Mobile communication systems as early as 1934 were based on Amplitude Modulation (AM) schemes and only certain public organizations maintained such systems. With the demand for newer and better mobile radio communication systems during the World War II and the development of Frequency Modulation (FM) technique by Edwin Armstrong, the mobile radio communication systems began to witness many new changes. Mobile telephone was introduced in the year 1946. However, during its initial three and a half decades it found very less market penetration owing to high costs and numerous technological drawbacks. But with the development of the cellular concept in the 1960s at the Bell Laboratories, mobile communications began to be a promising field of expanse which could serve wider populations. Initially, mobile communication was restricted to certain official users and the cellular concept was never even dreamt of being made commercially available. Moreover, even the growth in the cellular networks was very slow. However, with the development of newer and better technologies starting from the 1970s and with the mobile users now connected to the Public Switched Telephone Network (PSTN), there has been an astronomical growth in the cellular radio and the personal communication systems. Advanced Mobile Phone System (AMPS) was the first U.S. cellular telephone system and it was deployed in 1983. Wireless services have since then been experiencing a 50% per year growth rate. The number of cellular telephone users grew from 25000 in 1984 to around 3 billion in the year 2007 and the demand rate is increasing day by Day.

Mobile Telephony Development

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Cellular Concept:

The power of the radio signals transmitted by the BS decay as the signals travel away from it. A minimum amount of signal strength (let us say, x dB) is needed in order to be detected by the MS or mobile sets which may be the hand-held personal units or those installed in the vehicles. The region over which the signal strength lies above this threshold value x dB is known as the coverage area of a BS and it must be a circular region, considering the BS to be isotropic radiator. Such a circle, which gives this actual radio coverage, is called the footprint of a cell (in reality, it is amorphous). It might so happen that either there may be an overlap between any two such side by side circles or there might be a gap between the

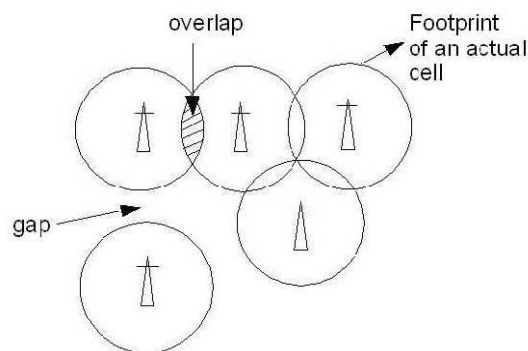


Fig 1:Footprint of cells showing the overlaps and gaps.

coverage areas of two adjacent circles. This is shown in Figure 1. Such a circular geometry, therefore, cannot serve as a regular shape to describe cells. We need a regular shape for cellular design over a territory which can be served by 3 regular polygons, namely, equilateral triangle, square and regular hexagon, which can cover the entire area without any overlap and gaps. Along with its regularity, a cell must be designed such that it is most reliable too, i.e., it supports even the weakest mobile with occurs at the edges of the cell. For any distance between the center and the farthest point in the cell from it, a regular hexagon covers the maximum area. Hence regular hexagonal geometry is used as the cells in mobile communication.

Frequency Reuse:

Frequency reuse, or, frequency planning, is a technique of reusing frequencies and channels within a communication system to improve capacity and spectral efficiency. Frequency reuse is one of the fundamental concepts on which commercial wireless systems are based that involve the partitioning of an RF radiating area into cells. The increased capacity in a commercial wireless network, compared with a network with a single transmitter, comes from the fact that the same radio frequency can be reused in a different area for a

completely different transmission. Frequency reuse in mobile cellular systems means that frequencies allocated to

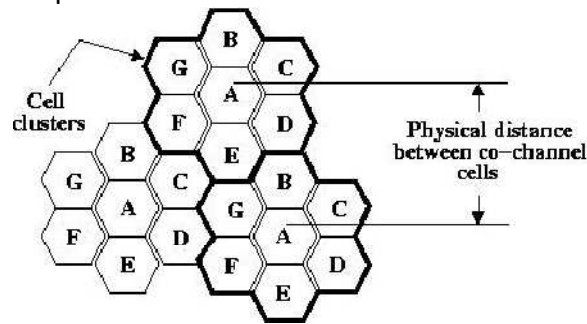


Fig 2: Frequency reuse technique of a cellular system.

the services are reused in a regular pattern of cells, each covered by one base station. The repeating regular pattern of cells is called cluster. Since each cell is designed to use radio frequencies only within its boundaries, the same frequencies can be reused in other cells not far away without interference, in another cluster. Such cells are called 'co-channel' cells. The reuse of frequencies enables a cellular system to handle a huge number of calls with a limited number of channels. Figure 3.2 shows a frequency planning with cluster size of 7, showing the co-channels cells in different clusters by the same letter. The closest distance between the co-channel cells (in different clusters) is determined by the choice of the cluster size and the layout of the cell cluster. Consider a cellular system with S duplex channels available for use and let N be the number of cells in a cluster. If each cell is allotted K duplex channels with all being allotted unique and disjoint channel groups we have $S = KN$ under normal circumstances. Now, if the cluster are repeated M times within the total area, the total number of duplex channels, or, the total number of users in the system would be $T = MS = KMN$. Clearly, if K and N remain constant, then

$T \propto M$ and, if T and K remain constant, then

$$N \propto \frac{1}{M}.$$

Hence the capacity gain achieved is directly proportional to the number of times a cluster is repeated, as shown in (3.1), as well as, for a fixed cell size, small N decreases the size of the cluster with in turn results in the increase of the number of clusters and hence the capacity. However for small N , co-channel cells are located much closer and hence more interference. The value of N is determined by calculating the amount of interference that can be tolerated for a sufficient quality communication. Hence the smallest N having interference below the tolerated limit is used. However, the cluster size N cannot take on any value and is given only by the following equation

$$N = i^2 + ij + j^2, \quad i \geq 0, j \geq 0,$$

Where i and j are integer numbers.

Channel Assignment Strategies

With the rapid increase in number of mobile users, the mobile service providers had to follow strategies which ensure the effective utilization of the limited radio spectrum. With

increased capacity and low interference being the prime objectives, a frequency reuse scheme was helpful in achieving these objectives. A variety of channel assignment strategies have been followed to aid these objectives. Channel assignment strategies are classified into two types: fixed and dynamic, as discussed below.

Fixed Channel Assignment (FCA)

In fixed channel assignment strategy each cell is allocated a fixed number of voice channels. Any communication within the cell can only be made with the designated unused channels of that particular cell. Suppose if all the channels are occupied, then the call is blocked and subscriber has to wait. This is simplest of the channel assignment strategies as it requires very simple circuitry but provides worst channel utilization. Later there was another approach in which the channels were borrowed from adjacent cell if all of its own designated channels were occupied. This was named as borrowing strategy. In such cases the MSC supervises the borrowing process and ensures that none of the calls in progress are interrupted.

Dynamic Channel Assignment (DCA)

In dynamic channel assignment strategy channels are temporarily assigned for use in cells for the duration of the call. Each time a call attempt is made from a cell the corresponding BS requests a channel from MSC. The MSC then allocates a channel to the requesting the BS. After the call is over the channel is returned and kept in a central pool. To avoid co-channel interference any channel that in use in one cell can only be reassigned simultaneously to another cell in the system if the distance between the two cells is larger than minimum reuse distance. When compared to the FCA, DCA has reduced the likelihood of blocking and even increased the trunking capacity of the network as all of the channels are available to all cells, i.e., good quality of service. But this type of assignment strategy results in heavy load on switching center at heavy traffic condition.

Handoff Process

When a user moves from one cell to the other, to keep the communication between the user pair, the user channel has to be shifted from one BS to the other without interrupting the call, i.e., when a MS moves into another cell, while the conversation is still in progress, the MSC automatically transfers the call to a new FDD channel without disturbing the conversation. This process is called as handoff. A schematic diagram of handoff is given in Figure Processing of handoff is an important task in any cellular system. Handoffs must be performed successfully and be imperceptible to the users. Once a signal

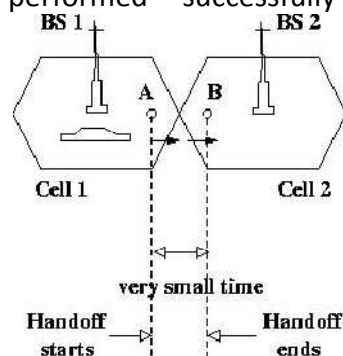


Fig 3: Handoff scenario at two adjacent cell boundary.

level is set as the minimum acceptable for good voice quality (P_{rmin}), then a slightly stronger level is chosen as the threshold (P_{rH}) at which handoff has to be made, as shown in Figure 3.4. A parameter, called power margin, defined as

$$\Delta = P_{rH} - P_{rmin}$$

is quite an important parameter during the handoff process since this margin can neither be too large nor too small. If Δ is too small, then there may not be enough time to complete the handoff and the call might be lost even if the user crosses the cell boundary. If Δ is too high on the other hand, then MSC has to be burdened with unnecessary handoffs. This is because MS may not intend to enter the other cell. Therefore Δ should be judiciously chosen to ensure imperceptible handoffs and to meet other objectives

Interference & System Capacity

Susceptibility and interference problems associated with mobile communications equipment are because of the problem of time congestion within the electromagnetic spectrum. It is the limiting factor in the performance of cellular systems. This interference can occur from clash with another mobile in the same cell or because of a call in the adjacent cell. There can be interference between the base stations operating at same frequency band or any other non-cellular system's energy leaking inadvertently into the frequency band of the cellular system. If there is an interference in the voice channels, cross talk is heard will appear as noise between the users. The interference in the control channels leads to missed and error calls because of digital signaling. Interference is more severe in urban areas because of the greater RF noise and greater density of mobiles and base stations. The interference can be

divided into 2 parts: co-channel interference and adjacent channel interference.

Co-channel interference (CCI)

For the efficient use of available spectrum, it is necessary to reuse frequency bandwidth over relatively small geographical areas. However, increasing frequency reuse also increases interference, which decreases system capacity and service quality. The cells where the same set of frequencies is used are called co-channel cells. Co-channel interference is the cross talk between two different radio transmitters using the same radio frequency as is the case with the co-channel cells. The reasons of CCI can be because of either adverse weather conditions or poor frequency planning or overly crowded radio spectrum. If the cell size and the power transmitted at the base stations are same then CCI will become independent of the transmitted power and will depend on radius of the cell (R) and the distance between the interfering co-channel cells (D). If D/R ratio is increased, then the effective distance between the co-channel cells will increase and interference will decrease. The parameter Q is called the frequency reuse ratio and is related to the cluster size. For hexagonal geometry

$$Q = D/R = \sqrt{3N}$$

From the above equation, small of ' Q ' means small value of cluster size ' N ' and increase in cellular capacity. But large ' Q ' leads to decrease in system capacity but increase in transmission quality. Choosing the options is very careful for the selection of ' N ', the proof of which is given in the first section. The Signal to Interference Ratio (SIR) for a mobile receiver which monitors the forward channel can be calculated as

$$\frac{S}{I} = \frac{S}{\sum_{i=1}^{i_0} I_i}$$

where i_0 is the number of co-channel interfering cells, S is the desired signal power from the baseband station and I_i is the interference power caused by the i -th interfering co-channel

base station. In order to solve this equation from power calculations, we need to look into the signal power characteristics. The average power in the mobile radio channel decays as a power law of the distance of separation between transmitter and receiver. The expression for the received power P_r at a distance d can be approximately calculated as

$$P_r = P_0 \left(\frac{d}{d_0} \right)^{-n}$$

and in the dB expression as

$$P_r(dB) = P_0(dB) - 10n \log\left(\frac{d}{d_0}\right)$$

where P_0 is the power received at a close-in reference point in the far field region at a small distance d_0 from the transmitting antenna, and 'n' is the path loss exponent. Let us calculate the SIR for this system. If D_i is the distance of the i-th interferer from the mobile, the received power at a given mobile due to i-th interfering cell is proportional to $(D_i)^{-n}$ (the value of 'n' varies between 2 and 4 in urban cellular systems). Let us take that the path loss exponent is same throughout the coverage area and the transmitted power be same, then SIR can be approximated as

$$\frac{S}{I} = \frac{R^{-n}}{\sum_{i=1}^{i_0} D_i^{-n}}$$

where the mobile is assumed to be located at R distance from the cell center. If we consider only the first layer of interfering cells and we assume that the interfering base stations are equidistant from the reference base station and the distance between the cell centers is 'D' then the above equation can be converted as

$$\frac{S}{I} = \frac{(D/R)^n}{i_0} = \frac{(\sqrt{3N})^n}{i_0}$$

which is an approximate measure of the SIR. Subjective tests performed on AMPS cellular system which uses FM and 30 kHz channels show that sufficient voice quality can be obtained by SIR being greater than or equal to 18 dB. If we take $n=4$, the value of 'N' can be calculated as 6.49. Therefore minimum N is 7. The above equations are based on hexagonal geometry and the distances from the closest interfering cells can vary if different frequency reuse plans are used. We can go for a more approximate calculation for co-channel SIR. This is the example of a 7 cell reuse case. The mobile is at a distance of $D-R$ from 2 closest interfering cells and approximately $D+R/2$, D , $D-R/2$ and $D+R$ distance from other interfering cells in the first tier. Taking $n = 4$ in the above equation, SIR can be approximately calculated as

$$\frac{S}{I} = \frac{R^{-4}}{2(D-R)^{-4} + (D+R)^{-4} + (D)^{-4} + (D+R/2)^{-4} + (D-R/2)^{-4}}$$

which can be rewritten in terms frequency reuse ratio Q as

$$\frac{S}{I} = \frac{1}{2(Q-1)^{-4} + (Q+1)^{-4} + (Q)^{-4} + (Q+1/2)^{-4} + (Q-1/2)^{-4}}$$

Using the value of N equal to 7 (this means $Q = 4.6$), the above expression yields that worst case SIR is 53.70 (17.3 dB). This shows that for a 7 cell reuse case the worst case SIR is slightly less than 18 dB. The worst case is when the mobile is at the corner of the cell i.e., on